

Packet Tracer - VLAN Configuration

Addressing Table

Device	Interface	IP Address	Subnet Mask	VLAN
PC1	NIC	172.17.10.21	255.255.255.0	10
PC2	NIC	172.17.20.22	255.255.255.0	20
PC3	NIC	172.17.30.23	255.255.255.0	30
PC4	NIC	172.17.10.24	255.255.255.0	10
PC5	NIC	172.17.20.25	255.255.255.0	20
PC6	NIC	172.17.30.26	255.255.255.0	30

Objectives

- Part 1: Verify the Default VLAN Configuration
- Part 2: Configure VLANs
- Part 3: Assign VLANs to Ports

Background

VLANs are helpful in the administration of logical groups, allowing members of a group to be easily moved, changed, or added. This activity focuses on creating and naming VLANs, and assigning access ports to specific VLANs.

Part 1: View the Default VLAN Configuration

Step 1: Display the current VLANs.

On S1, issue the command that displays all VLANs configured. By default, all interfaces are assigned to VLAN 1.

Step 2: Verify connectivity between PCs on the same network.

Notice that each PC can ping the other PC that shares the same subnet.

- = PC1 can ping PC4
- = PC2 can ping PC5
- = PC3 can ping PC6

Pings to hosts on other networks fail.

- What benefits can VLANs provide to the network?
- **ความปลอดภัย (Security):** แยกกลุ่มข้อมูลสำคัญออกจากส่วนอื่นของเครือข่าย
- **ประหยัดค่าใช้จ่าย (Cost Reduction):** แบ่งวงเครือข่ายได้โดยไม่ต้องซื้อสวิตช์เพิ่ม
- **ประสิทธิภาพสูงขึ้น (Better Performance):** จำกัดวงของทราฟฟิก Broadcast ทำให้อินเทอร์เน็ตไม่หน่วง
- **ง่ายต่อการจัดการ (Easier Management):** ผู้ดูแลระบบสามารถจัดกลุ่มผู้ใช้งานตามแผนกหรือหน้าที่ได้ง่ายขึ้น

Part 2: Configure VLANs

Step 1: Create and name VLANs on S1.

a. Create the following VLANs. Names are case-sensitive and must match the requirement exactly:

- o VLAN 10: Faculty/Staff

```
S1#(config)# vlan 10
```

```
S1#(config-vlan)# name Faculty/Staff
```

b. Create the remaining VLANs.

- o VLAN 20: Students
- o VLAN 30: Guest(Default)
- o VLAN 99: Management&Native
- o VLAN 150: VOICE

Step 2: Verify the VLAN configuration.

Which command will only display the VLAN name, status, and associated ports on a switch?

Step 3: Create the VLANs on S2 and S3.

Use the same commands from Step 1 to create and name the same VLANs on S2 and S3.

Step 4: Verify the VLAN configuration.

Part 3: Assign VLANs to Ports

Step 1: Assign VLANs to the active ports on S2.

a. Configure the interfaces as access ports and assign the VLANs as follows:

- o VLAN 10: FastEthernet 0/11

```
S2(config)# interface f0/11
```

```
S2(config-if)# switchport mode access
```

```
S2(config-if)# switchport access vlan 10
```

b. Assign the remaining ports to the appropriate VLAN.

- o VLAN 20: FastEthernet 0/18
- o VLAN 30: FastEthernet 0/6

Step 2: Assign VLANs to the active ports on S3.

S3 uses the same VLAN access port assignments as S2. Configure the interfaces as access ports and assign the VLANs as follows:

- o VLAN 10: FastEthernet 0/11
- o VLAN 20: FastEthernet 0/18
- o VLAN 30: FastEthernet 0/6

Step 3: Assign the VOICE VLAN to FastEthernet 0/11 on S3.

As shown in the topology, the S3 FastEthernet 0/11 interface connects to a Cisco IP Phone and PC4. The IP phone contains an integrated three-port 10/100 switch. One port on the phone is labeled Switch and connects to F0/4. Another port on the phone is labeled PC and connects to PC4. The IP phone also has an internal port that connects to the IP phone functions.

The S3 F0/11 interface must be configured to support user traffic to PC4 using VLAN 10 and voice traffic to the IP phone using VLAN 150. The interface must also enable QoS and trust the Class of Service (CoS) values assigned by the IP phone. IP voice traffic requires a minimum amount of throughput to

support acceptable voice communication quality. This command helps the switchport to provide this minimum amount of throughput.

```
S3(config)# interface f0/11
S3(config-if)# mls qos trust cos
S3(config-if)# switchport voice vlan 150
```

Step 4: Verify loss of connectivity.

Previously, PCs that shared the same network could ping each other successfully.

Study the output of from the following command on **S2** and answer the following questions based on your knowledge of communication between VLANs. Pay close attention to the Gig0/1 port assignment.

```
S2# show vlan brief
```

```
VLAN Name Status Ports
```

```
-----
1 default active Fa0/1, Fa0/2, Fa0/3, Fa0/4
Fa0/5, Fa0/7, Fa0/8, Fa0/9
Fa0/10, Fa0/12, Fa0/13, Fa0/14
Fa0/15, Fa0/16, Fa0/17, Fa0/19
Fa0/20, Fa0/21, Fa0/22, Fa0/23
Fa0/24, Gig0/1, Gig0/2
10 Faculty/Staff active Fa0/11
20 Students active Fa0/18
30 Guest(Default) active Fa0/6
99 Management&Native active
150 VOICE active
```

Try pinging between PC1 and PC4.

Although the access ports are assigned to the appropriate VLANs, were the pings successful? Explain.

คำตอบ: ไม่สำเร็จ เนื่องจากพอร์ตหรือสายสัญญาณที่เชื่อมต่อระหว่างสลับสายสวิตช์ (พอร์ต Gig0/1) ยังคงทำงานอยู่ในโหมด Access และเป็นสมาชิกของ VLAN 1 อยู่ตามค่าเริ่มต้น ซึ่งพอร์ตแบบ Access จะสามารถส่งข้อมูลได้เพียงแค่ VLAN เดียวเท่านั้น ดังนั้น เมื่อ PC1 พยายามส่งแพ็กเก็ตข้อมูลของ VLAN 10 ข้ามไปยังสวิตช์อีกฝั่งเพื่อหา PC4 แพ็กเก็ตนั้นจึงถูกบล็อกหรือถูกทิ้งไป ทำให้ ping ไม่สำเร็จ

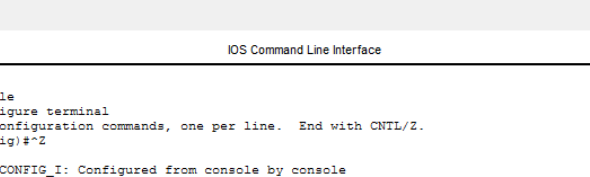
What could be done to resolve this issue?

คำตอบ: วิธีแก้ไขคือ จะต้องเปลี่ยนโหมดพอร์ตที่เชื่อมต่อระหว่างสวิตช์ทั้งหมด (เช่น พอร์ต **Gig0/1** บน S2, S3 และพอร์ตบน S1) ให้เป็น **Trunk Port** โดยใช้คำสั่ง:

None

switchport mode trunk

การตั้งค่าเป็นพอร์ต Trunk จะช่วยให้สายเชื่อมต่อระหว่างสวิตช์สามารถแนบป้ายกำกับ (Tag) และรับ-ส่งข้อมูลของหลาย ๆ VLAN (ทั้ง VLAN 10, 20, 30) ข้ามไปหาอีกสวิตช์ได้สำเร็จ



The screenshot shows a Cisco IOS CLI window titled "IOS Command Line Interface". The user has entered the following commands:

```

S1>enable
S1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#^Z
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#show vlan brief

```

The output of the `show vlan brief` command is displayed as a table:

VLAN Name	Status	Ports
1 default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

The prompt returns to `S1#`.

Logical Physical x: 585, y: 377
Root
05:15:30

Part 3: Assign VLANs to Ports

Background

VLANs are helpful in the administration of logical groups, allowing members of a group to be easily moved, changed, or added. This activity focuses on creating and naming VLANs, and assigning access ports to specific VLANs.

Part 1: View the Default VLAN Configuration

Step 1: Display the current VLAN configuration

On S1, issue the command that displays the current VLAN configuration. By default, all interfaces are assigned to the default VLAN.

Step 2: Verify connectivity between network.

Notice that each PC can ping the other PC on the same subnet.

- PC1 can ping PC4
- PC2 can ping PC5
- PC3 can ping PC6

Pings to hosts on other networks fail.

What benefits can VLANs provide?

Part 2: Configure VLANs

Step 1: Create and name VLANs

- Create the following VLANs. Names must match the requirements.
 - VLAN 10: Faculty/Staff
 - VLAN 20: Students
 - VLAN 30: Guest(Default)

CLI

IOS Command Line Interface

```

1002 fddi-default          active
1003 token-ring-default   active
1004 fddinet-default      active
1005 trnet-default        active
S1#conf
S1#configure ter
S1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#vlan 10
S1(config-vlan)#name Faculty/Staff
S1(config-vlan)#vlan 20
S1(config-vlan)#name Students
S1(config-vlan)#vlan 30
S1(config-vlan)#name Guest(Default)
S1(config-vlan)#vlan 99
S1(config-vlan)#name Management&Native
S1(config-vlan)#vlan 150
S1(config-vlan)#name VOICE
S1(config-vlan)#end
S1#
*SYS-5-CONFIG_I: Configured from console by console
            
```

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Time Elapsed: 00:11:12
Completion: 22%

✓ Dock
Check Results
Back
1/1
Next

LogicalPhysical

- PC3 can ping PC6
- Pings to hosts on other VLANs
- What benefits can VLANs provide?

Part 2: Configure VLANs on S2

Step 1: Create and name the VLANs

- Create the following VLANs on S2. The names must match the requirements.
 - VLAN 10: Faculty/Staff
- Create the remaining VLANs on S2.
 - VLAN 20: Students
 - VLAN 30: Guest (Default)
 - VLAN 99: Management & Native
 - VLAN 150: VOICE

Step 2: Verify the VLAN configuration

Which command will you use to verify the configuration of the VLANs on S2?

Step 3: Create the VLANs on S1

Use the same commands as you used on S2 to create the same VLANs on S1.

Step 4: Verify the VLAN configuration on S1

Part 3: Assign VLANs to the interfaces

Step 1: Assign VLANs to the interfaces

- Configure the interfaces on S2 as follows:
 - VLAN 10: FastEthernet 0/1, 0/2, 0/3, 0/4, 0/5, 0/6, 0/7, 0/8, 0/9, 0/10, 0/11, 0/12, 0/13, 0/14, 0/15, 0/16, 0/17, 0/18, 0/19, 0/20, 0/21, 0/22, 0/23, 0/24

CLI

IOS Command Line Interface

```
S2>en
S2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
S2(config)#vlan 10
S2(config-vlan)#name Faculty/Staff
S2(config-vlan)#vlan 20
S2(config-vlan)#name Students
S2(config-vlan)#vlan 30
S2(config-vlan)#name Guest(Default)
S2(config-vlan)#vlan 99
S2(config-vlan)#name Management&Native
S2(config-vlan)#vlan 150
S2(config-vlan)#name VOICE
S2(config-vlan)#end
S2#
%SYS-5-CONFIG_I: Configured from console by console

S2#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4,
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8,
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12,
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16,
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20,
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2
10   Faculty/Staff           active
20   Students                active
30   Guest(Default)          active
99   Management&Native       active
150  VOICE                   active
1002 fddi-default            active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active
S2#
```

Copy

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☐ Top

Time Elapsed: 00:14:46

Completion: 45%

☒ Dock

Check Results

Back

1/1

Next

PC4
172.17.10.24
VLAN10

PC5
172.17.20.25
VLAN20

PC6
172.17.30.26
VLAN30

LogicalPhysicalx: 253

08:05:00

• PC3 can ping PC6

Pings to hosts on other network

What benefits can VLANs provide?

Part 2: Configure VLANs

Step 1: Create and name VLANs

a. Create the following VLANs on S3. The names must match the requirements.

o VLAN 10: Faculty/Staff

S3(config)# vlan 10

S3(config-vlan)# name Faculty/Staff

S3(config-vlan)# vlan 20

S3(config-vlan)# name Students

S3(config-vlan)# vlan 30

S3(config-vlan)# name Guest(Default)

S3(config-vlan)# vlan 99

S3(config-vlan)# name Management&Native

S3(config-vlan)# vlan 150

S3(config-vlan)# name VOICE

S3(config-vlan)# end

S3#

!SYS-5-CONFIG_I: Configured from console by console

b. Create the remaining VLANs on S1 and S2.

o VLAN 20: Students

o VLAN 30: Guest(Default)

o VLAN 99: Management&Native

o VLAN 150: VOICE

Step 2: Verify the VLAN configuration

Which command will only display the ports associated with a specific VLAN?

Step 3: Create the VLANs on S1 and S2

Use the same commands to create the same VLANs on S2 and S3.

Step 4: Verify the VLAN configuration

Part 3: Assign VLANs to PC interfaces

Step 1: Assign VLANs to the PC interfaces

a. Configure the interfaces on PC3 to be in the following VLANs:

o VLAN 10: FastEthernet0/24

CLI

IOS Command Line Interface

S3>en

S3#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

S3(config)#vlan 10

S3(config-vlan)#name Faculty/Staff

S3(config-vlan)#vlan 20

S3(config-vlan)#name Students

S3(config-vlan)#vlan 30

S3(config-vlan)#name Guest(Default)

S3(config-vlan)#vlan 99

S3(config-vlan)#name Management&Native

S3(config-vlan)#vlan 150

S3(config-vlan)#name VOICE

S3(config-vlan)#end

S3#

!SYS-5-CONFIG_I: Configured from console by console

S3#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	Faculty/Staff	active	
20	Students	active	
30	Guest(Default)	active	
99	Management&Native	active	
150	VOICE	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

S3#

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Top

Time Elapsed: 00:16:53

Completion: 68%

✓ Dock

Check Results

Back

1/1

Next

Logical **Physical** x: 335, y: 237

- VLAN 30: Guest(Default)
- VLAN 99: Management&Native
- VLAN 150: VOICE

Step 2: Verify the VLAN configuration.

Which command will only display the VLAN name, status, and associated ports on a switch?

Step 3: Create the VLANs on S2 and S3.

Use the same command to create the same VLANs on S2 and S3.

Step 4: Verify the VLAN configuration on S2 and S3.

Part 3: Assign VLANs to the interfaces.

Step 1: Assign VLANs to the interfaces on S2.

- Configure the interfaces on S2 for the VLANs as follows:
 - VLAN 10: FastEthernet 0/11
 - VLAN 20: FastEthernet 0/18
 - VLAN 30: FastEthernet 0/6
- Assign the remaining interfaces on S2 to the default VLAN (VLAN 1).

CLI

IOS Command Line Interface

```

S2#en
S2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
S2(config)#interface fastethernet 0/11
S2(config-if)#switchport mode access
S2(config-if)#switchport access vlan 10
S2(config-if)#exit
S2(config)#interface fastethernet 0/18
S2(config-if)#switchport mode access
S2(config-if)#switchport access vlan 20
S2(config-if)#exit
S2(config)#interface fastethernet 0/6
S2(config-if)#switchport mode access
S2(config-if)#switchport access vlan 30
S2(config-if)#end
S2#
%SYS-5-CONFIG_I: Configured from console by console

S2#write
Building configuration...
[OK]
S2#
S2#
            
```

Copy Paste

Step 2: Assign VLANs to the interfaces on S3.

S3 uses the same VLAN configuration as S2. Configure the interfaces on S3 for the VLANs as follows:

- VLAN 10: FastEthernet 0/11
- VLAN 20: FastEthernet 0/18
- VLAN 30: FastEthernet 0/6

Time Elapsed: 00:20:46

Completion: 81%

✓ Dock
Check Results
Back
1/1
Next

Root
↶
↷
↻
👤
🌐
10:01:00

Logical Physical x: 408, y: 958
Root
11:38:00

using VLAN 150. The interface must also enable QoS and trust the Class of Service (CoS) values assigned by the IP phone. IP voice traffic requires a minimum amount of throughput to support acceptable voice communication quality. This command helps the switchport to provide this minimum amount of throughput.

```

S3(config)# interface f0/11
S3(config-if)# mls qos trust cos
S3(config-if)# switchport voice vlan 150

```

Step 4: Verify loss of conn

Previously, PCs that shared other successfully.

Study the output of from the answer the following question communication between VLAN Gig0/1 port assignment.

```

S2# show vlan bri
VLAN Name Status Po
-----
1 default active Fa
Fa0/5, Fa0/7, Fa0/8
Fa0/10, Fa0/12, Fa0
Fa0/15, Fa0/16, Fa0
Fa0/20, Fa0/21, Fa0
Fa0/24, Gig0/1, Gig
10 Faculty/Staff ac
20 Students active
30 Guest(Default) a
99 Management/Nativ
150 VOICE active

```

Try pinging between PC1 and PC4.

Although the access ports are in different VLANs, were the pings successful?

What could be done to resolve this issue?

S3

CLI

IOS Command Line Interface

```

S3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
S3(config)#interface fastethernet 0/11
S3(config-if)#switchport mode access
S3(config-if)#switchport access vlan 10
S3(config-if)#exit
S3(config)#interface fastethernet 0/18
S3(config-if)#switchport mode access
S3(config-if)#switchport access vlan 20
S3(config-if)#exit
S3(config)#interface fastethernet 0/6
S3(config-if)#switchport mode access
S3(config-if)#switchport access vlan 30
S3(config-if)#interface fastethernet 0/11
S3(config-if)#mls qos trust cos
S3(config-if)#switchport voice vlan 150
S3(config-if)#end
S3#
%SYS-5-CONFIG_I: Configured from console by console

S3#write
Building configuration...
[OK]
S3#

```

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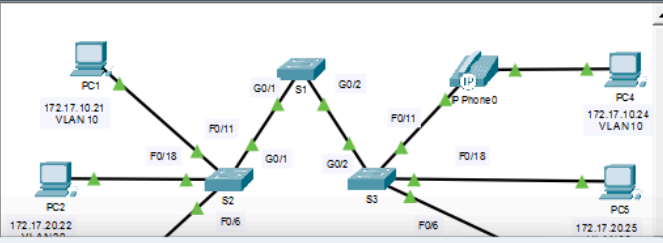
Time Elapsed: 00:24:00
Completion: 100%

☒ Dock
Check Results
Back
1/1
Next

LogicalPhysicalx: 704, y: 929Root12:18:30

using VLAN 150. The interface must also enable QoS and trust the Class of Service (CoS) values assigned by the IP phone. IP voice traffic requires a minimum amount of throughput to support acceptable voice communication quality. This command helps the switchport to provide this minimum amount of throughput.

S3(config)# interface f0/11
S3(config-if)# mls qos trust cos
S3(config-if)# switchport voice vlan 150



Step 4: Verify loss of connectivity.

Previously, PCs that shared the same other successfully.

Study the output of from the following answer the following questions based communication between VLANs. Pay Gig0/1 port assignment.

```
S2# show vlan brief
VLAN Name Status Ports
-----
1 default active Fa0/1, Fa0/5, Fa0/7, Fa0/8, Fa0/9, Fa0/10, Fa0/12, Fa0/13, Fa0/15, Fa0/16, Fa0/17, Fa0/20, Fa0/21, Fa0/22, Fa0/24, Gig0/1, Gig0/2
10 Faculty/Staff active Fa0/18
20 Students active Fa0/18
30 Guest(Default) active Fa
99 ManagementNative active
150 VOICE active
```

Try pinging between PC1 and PC4.

Although the access ports are assign VLANs, were the pings successful? E

What could be done to resolve this is

PC1

DesktopProgramming

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ping 172.17.10.24

Pinging 172.17.10.24 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.17.10.24:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

Top

Time Elapsed: 00:25:22

Completion: 100%

Back1/1Next

Activity Results

Time Elapsed: 00:32:20

Congratulations Guest! You completed the activity.

Overall Feedback **Assessment Items** Connectivity Tests

Expand/Collapse All

Show Incorrect Items

Score : 94/94
Item Count : 22/22

Assessment Items	Status	Points	Component(s)	Feedback
[-] Network				
[-] S1				
[-] VLANS				
[-] VLAN 10		0	Switching	
[-] ✓ VLAN Name	Correct	5	VLAN Configurat...	
[-] VLAN 150		1	Switching	
[-] ✓ VLAN Name	Correct	4	Switching	
[-] VLAN 20		0	Switching	
[-] ✓ VLAN Name	Correct	4	VLAN Configurat...	
[-] VLAN 30		0	Switching	
[-] ✓ VLAN Name	Correct	4	VLAN Configurat...	
[-] VLAN 99		0	Switching	
[-] ✓ VLAN Name	Correct	4	VLAN Configurat...	
[-] S2				
[-] Ports				
[-] FastEthernet0/11		0	Other	
[-] ✓ Access VLAN	Correct	5	VLAN Configurat...	
[-] FastEthernet0/18		0	Other	
[-] ✓ Access VLAN	Correct	5	VLAN Configurat...	
[-] FastEthernet0/6		0	Other	
[-] ✓ Access VLAN	Correct	5	VLAN Configurat...	
[-] VLANS				
[-] VLAN 10		0	Switching	
[-] ✓ VLAN Name	Correct	5	VLAN Configurat...	
[-] VLAN 150		1	Switching	
[-] ✓ VLAN Name	Correct	4	Switching	
[-] VLAN 20		0	Switching	
[-] ✓ VLAN Name	Correct	4	VLAN Configurat...	
[-] VLAN 30		0	Switching	
[-] ✓ VLAN Name	Correct	4	VLAN Configurat...	
[-] VLAN 99		0	Switching	
[-] ✓ VLAN Name	Correct	4	VLAN Configurat...	
[-] S3				
[-] Ports				
[-] FastEthernet0/11				
[-] ✓ Access VLAN	Correct	4	VLAN Configurat...	
[-] ✓ Voice VLAN	Correct	4	Switching	
[-] FastEthernet0/18		0	Other	
[-] ✓ Access VLAN	Correct	4	VLAN Configurat...	
[-] FastEthernet0/6		0	Other	
[-] ✓ Access VLAN	Correct	4	VLAN Configurat...	
[-] VLANS				
[-] VLAN 10		0	Switching	
[-] ✓ VLAN Name	Correct	5	VLAN Configurat...	
[-] VLAN 150		1	Switching	
[-] ✓ VLAN Name	Correct	4	VLAN Configurat...	
[-] VLAN 20		0	Switching	
[-] ✓ VLAN Name	Correct	4	VLAN Configurat...	
[-] VLAN 30		0	Switching	
[-] ✓ VLAN Name	Correct	4	VLAN Configurat...	
[-] VLAN 99		0	Switching	
[-] ✓ VLAN Name	Correct	4	VLAN Configurat...	

Component	Items/Total	Score
Switching	3/3	12/12
VLAN Configuration	19/19	82/82

Close